Population shifts in Finland

LAB University of Applied Sciences
Bachelor's Degree Programme in Industrial Information Technology
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The author of this thesis, John Doe, is responsible for the accuracy of the entire contents of the thesis, including content generated with AI. AI was used in this work during background research and later for language editing.

Background research made use of Copilot (Microsoft Copilot: Your everyday AI companion).

The following prompts were entered on August 28, 2025:

How does a heat pump work?

The generative AI output was adapted and edited for the final answer. Original output has been saved and is available upon request.

DeepL (DeepL Translate: The world's most accurate translator) was used for language checking to ensure linguistic correctness. The use was occasional throughout the thesis.

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Abstract

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Title of the thesis

Population shifts in Finland

Degree, Field of Study

Industrial Information Technology

Name, title and organisation of the client

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Abstract

Type the abstract text here …

Keywords

keyword, keyword, keyword

1. Executive Summary

This report analyzes the demographic trajectory of Finland using historical data from 1972 to 2024. The primary objective is to evaluate the impact of recent structural breaks on future population forecasts. The findings suggest that Finland has entered a new demographic regime where external migration has become the sole driver of population growth, compensating for a record-low natural birth rate.

2. Methodology

The data was retrieved directly from the Statistics Finland database via the PxWeb API.

- Data Source: Population structure by area.
- Processing: Statistical anomaly detection was performed using a 2-sigma threshold to identify significant deviations from historical trends.
- Modeling: Iterative scenario modeling was applied to project population counts up to the year 2050.

3. Anomaly Detection

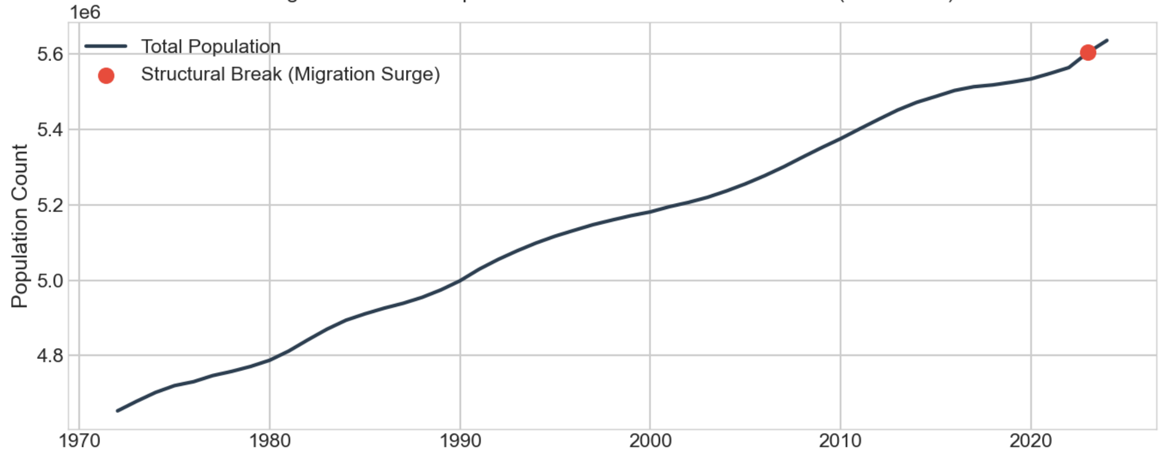
The analysis identified a significant Structural Anomaly in 2023.

- Historical average growth: ~15,000 people per year.
- 2023 observed growth: ~40,000 people.
- 2024 observed growth: ~32,000 people.

Analysis.

This represents a "Turning Point." In previous decades, population growth was driven by a balance of births and moderate migration. Currently, the "Post-Anomaly" trend shows that migration surge is the primary factor maintaining Finland's population above 5.6 million.

Figure 1: Finland Population Growth & Identified Anomalies (1972-2024)

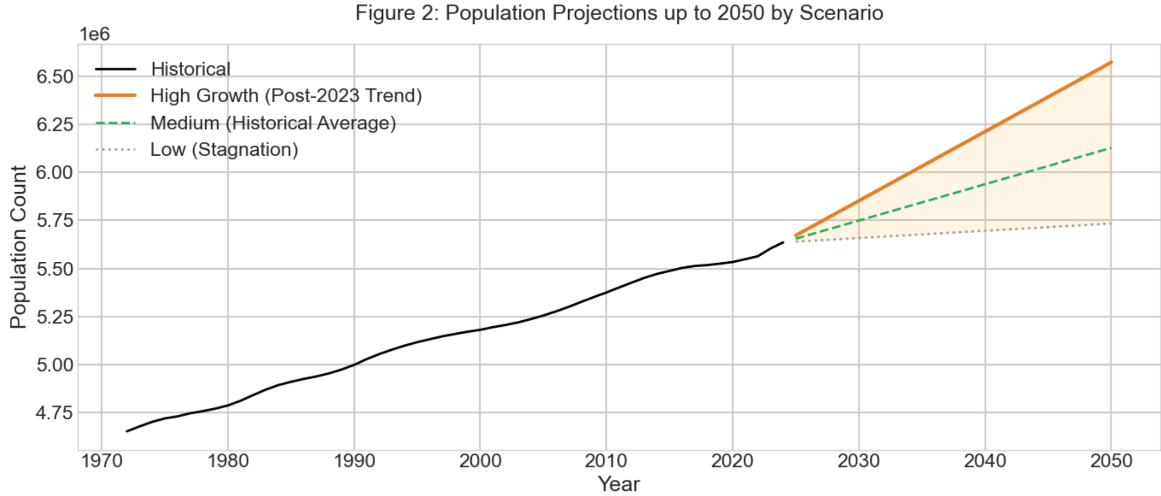


4. Forecasting Scenarios to 2050

Based on the identified anomalies, we have developed three distinct scenarios for Finland's future:

Scenario	Description	Projected 2050 Population
High Growth	Assumes the 2023-2024 migration surge becomes the new baseline.	~6,440,000
Medium Trend	Assumes a return to long-term historical averages.	~5,960,000
Stagnation	Assumes reduced migration and continued birth rate decline.	~5,680,000

Figure 2: Population Projections up to 2050 by Scenario



Part 5: Conclusion

The record-breaking migration seen in 2023-2024 has fundamentally shifted the demographic outlook for Finland. Traditional linear projections that do not account for this structural break will likely underestimate future needs for housing, healthcare, and infrastructure. Policymakers should consider the high growth scenario as a plausible future, given current labor market demands.